InHand Networks

Compact Edge Gateway, Reliable Connectivity, Industrial Design



IG101 Edge Gateway

- LTE Cat.1
- Modbus to MQTT
- Edge Computing
- Device Manager

1. Product Overview

IG101 is a compact industrial edge gateway for IIoT applications, delivering cost-effective LTE connectivity and flexible edge data processing for large-scale device networking.

Product Highlights:

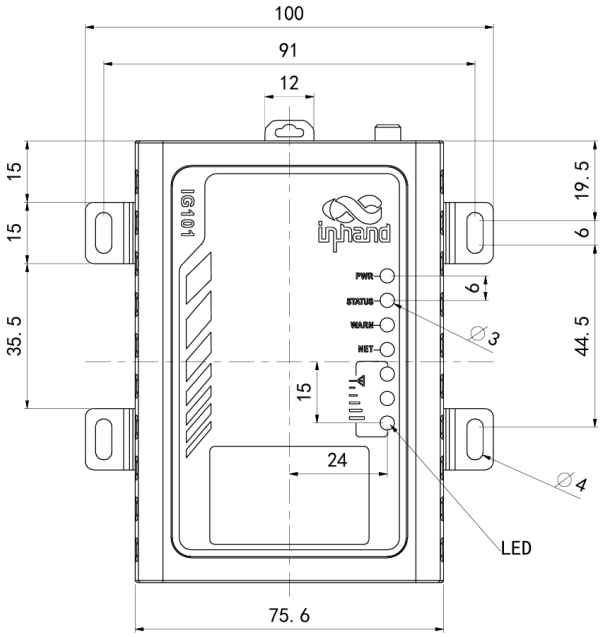
- **Reliable Cellular Access:** LTE Cat.1 connectivity for always-on remote access
- **Edge Data Processing:** Supports customized edge computing and data preprocessing
- **Industrial Reliability:** Metal housing, watchdog, and multi-level link detection
- **Cloud Integration:** Supports MQTT and fast integration with third-party cloud platforms
- **Scalable O&M:** Supports InHand Device Manager for large-scale deployment and management

Core Technical Specifications

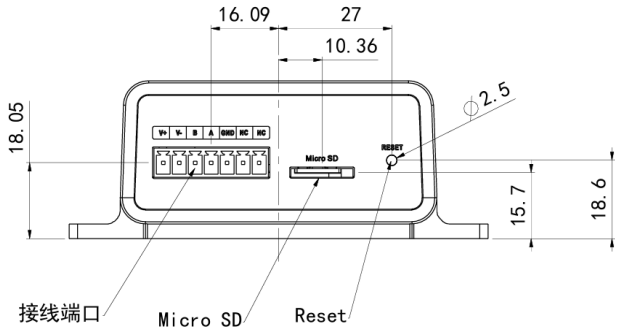
Item	Specification
Cellular Network	LTE Cat.1
Network Access and Authentication	APN, VPDN; CHAP/PAP
Industrial Protocol	Modbus RTU
Cloud Connectivity	Standard MQTT protocol, connects to third-party cloud platforms
Remote Management	InHand Device Manager Platform
CPU	ARM Cortex-A5

Item	Specification
Memory and Flash	4 MB RAM + 8 MB Flash
Interface Set	1 x RS-485, 1 x RS-232, 1 x SIM, 7-PIN terminal
Power and Consumption	DC 7-38V; standby 150 mA @ 12V, working 280 mA @ 12V, peak 340 mA @ 12V
Operating Temperature	-20 °C ~ 70 °C (-4 °F ~ 158 °F)
Dimensions and Protection	76 × 108 × 37.5 mm (2.99 × 4.25 × 1.48 in), IP30

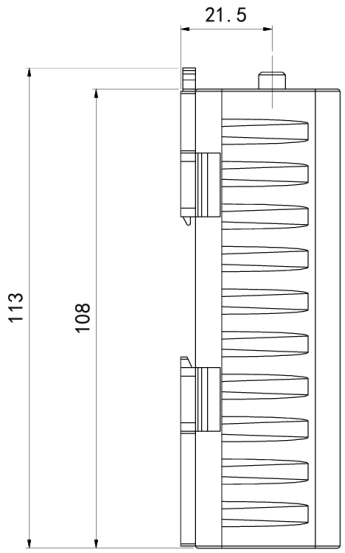
2. Product Dimensions & Terminal Definition



Top View



Interface Side



Side View

Notes:
1. All dimensions are in millimeters (mm), with inches (in) in parentheses.
2. All dimensions are approximate and for reference only.
3. Drawing dimensions are not intended for manufacturing.
4. Dimensions are subject to part and process tolerances.
5. Specifications are subject to change without notice.

7 PIN Definition

PIN	Definition	Description
1	V+	Positive electrode
2	V-	Negative electrode
3	B	RS-485- of serial 1
4	A	RS-485+ of serial 1
5	GND	Signal ground
6	TXD	Serial RS-232 send
7	RXD	Serial RS-232 receive

3. Hardware Specifications

Category/Parameter	Specification
CPU and Storage	
CPU	ARM Cortex-A5

Category/Parameter	Specification
RAM	4 MB
Flash	8 MB
Connectivity and Interfaces	
Serial Ports	1 x RS-485, 1 x RS-232, industrial terminal block
SIM Card Slot	1 x SIM, 1.8V/3V
Reset Button	Pinhole reset button
LED Indicators	PWR, NET, STATUS, WARN, Signal Strength (3)
microSD card Slot (microSD)	microSD, up to 32 GB
Power and Power Consumption	
Power Input	DC 7-38V, reverse polarity protection
Power Interface	Industrial terminal block
Standby Power	150 mA @ 12V
Working Power	280 mA @ 12V
Peak Power	340 mA @ 12V
Mechanical Specifications	
Dimensions (L x W x H)	76 × 108 × 37.5 mm (2.99 × 4.25 × 1.48 in)
Mounting Method	DIN-rail, wall mounting
Protection Rating	IP30

Category/Parameter	Specification
Housin	Metal housing
Hardware Watchdog	Device self-diagnosis and auto-recovery (fuzzy-mapped from Embedded Watchdog)
Environment and Certifications	
Storage Temperature	-40 ~ 85 C (-40 °F ~ 185 °F)
Operating Temperature	-20 ~ 70 C (-4 °F ~ 158 °F)
Ambient Humidity	5 ~ 95% (non-condensing)
Physical Characteristics	IEC60068-2-27 shock resistance; IEC60068-2-6 vibration resistance; IEC60068-2-32 drop resistance
EMC Standard	EN61000-4-2, level 3, Static EN61000-4-3, level 3, Radiation Electric Field EN61000-4-4, level 3, Pulsed Electric Field EN61000-4-5, level 3, Surge EN61000-4-6, level 3, Conducted Disturbance Immunity EN61000-4-8, Power Frequency Field Resistance, horizontal / vertical 400A/m (>level 3) EN61000-4-12, level 3, Shock Wave Resistance

4. Software Specifications

Category/Parameter	Specification
Operating System	
OS	FreeRTOS
Network Features	

Category/Parameter	Specification
Network Access	APN, VPDN
Access Authentication	CHAP/PAP
Network Types/Standards	LTE Cat.1
IP Applications	ICMP, DNS, TCP/UDP, TCP Server
Security	
User Management	Multi-level user authorization
AAA (Authentication, Authorization, Accounting)	CHAP/PAP
Reliability	
Link Detection/Probing	Heartbeat packet detection, auto-redial after disconnection
Embedded Watchdog	Device self-diagnosis and auto-recovery
Open Platform and Data Acquisition Protocols (DSA)	
Python Secondary Development	Customization development platform
DSA Protocol Engine	Standard MQTT protocol, connects to third-party cloud platforms
Industrial Protocols	Modbus RTU
Network Management	

Category/Parameter	Specification
Configuration Methods	Local/remote configuration tools
Upgrade Methods	Local or remote firmware upgrade
Log Function	Local logs, remote logs, important log power-off preservation
Configuration Backup	Import/export configuration files
Remote Management	InHand Device Manager Platform
Network Management Functions	InHand Device Manager Platform
Network Diagnostics	Ping

5. Ordering Information

Model Rule

Model code: IG101-<WMNN>-<IO/NA>

<WMNN>: WLAN & MODULE

Model List

Model	Region	<WMNN>: WLAN & MODULE	<IO/NA>
IG101-LQA3-STD	China	LTE-FDD: B1/B3/B5/B8 LTE-TDD: B34/B38/B39/B40/B41 GSM: B3/B5/B8	1 x RS-485 + 1 x RS-232, no I/O

IG101-FQ53-STD	EMEA	LTE-FDD: B1/B3/B5/B7/B8/B20/B28 LTE-TDD: B38/B40/B41 GSM: B2/B3/B5/B8	1 x RS-485 + 1 x RS-232, no I/O

6. Contact Us

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